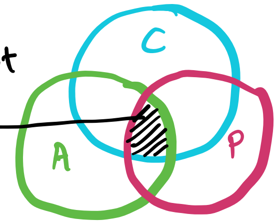
CAP Theorem

↓

Relates to DBs or distributed DBs

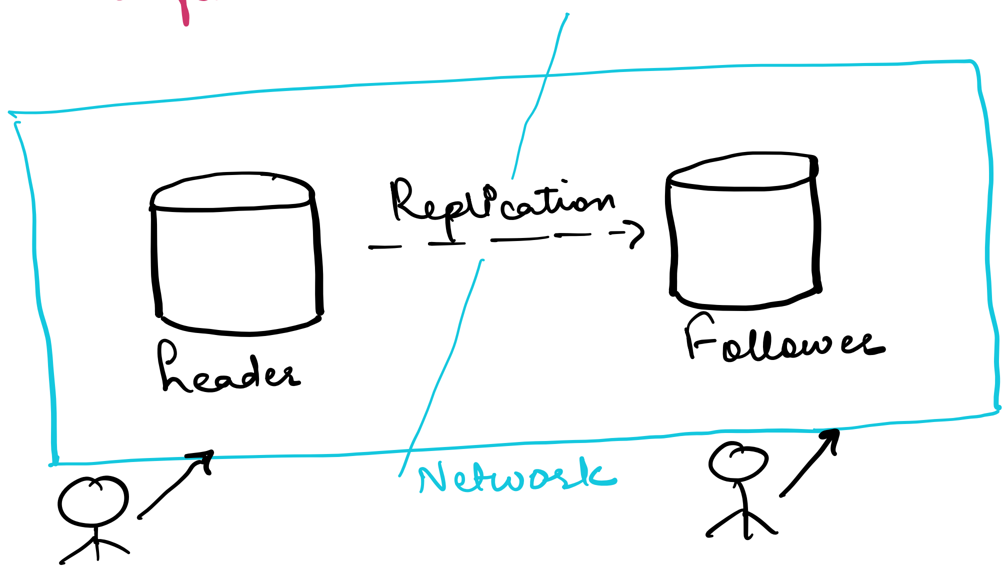
This cannot be possible



Either { C - Consistency
A - Availability

2/3 = Gauranteed to be use → P - Partition Tolerance

✗ CAP Theorem mainly applies when we replicate the data so mostly applies to NoSQL DBs as it has replication by default.



1) If the system with 2 nodes gets partitioned/cut then 2 nodes/dBs can't talk to eachother but Partition Tolerance means eventhough the 2 nodes get split or system gets disconnected still the system will continue to function

2) Consistency here means data consistency between the replicated data so every single read will get the most up to date data

Hence only

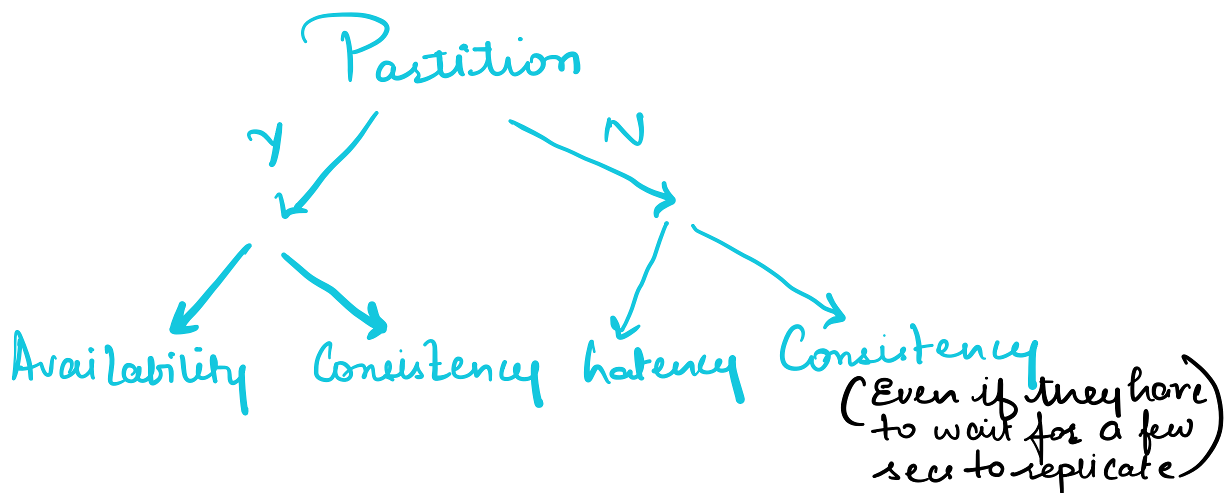
1 of them is achievable

but incase of network partition this might no hold true

3) Availability means that every user that did not crash will be avaible to respond to the user request even if some of the nodes have stale data

PACELC : (Extension of CAP Theorem)

Given P, choose A or C.
Else, favor Latency or Consistency



* This shows the tradeoffs we need to make while designing a distribute database system